MODULE 5

From Angles and Distance to X, Y, Z

5.1 The idea

SIMPLE APPROACH

To tell a friend where a treasure is hidden, you could say: "Face the big tree. Turn a little to the left. Look a little up. Now walk 25 steps that way." That is a **turn**, a **tilt** and a **distance**. A LiDAR records the same three things for every dot. Then it does some triangle math to change them into "how far forward, how far sideways, how far up" — which we call X, Y and Z.

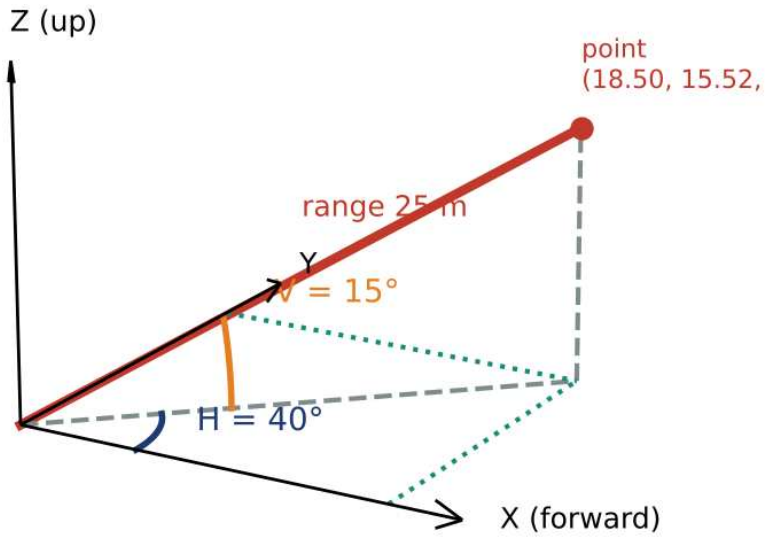


Figure 5.1: A range of 25 m with horizontal angle 40° and vertical angle 15° becomes a point at (18.50, 15.52, 6.47) m.

KEY EQUATION — Polar (spherical) to Cartesian

$$x_s = \rho \cos V \cos H, \qquad y_s = \rho \cos V \sin H, \qquad z_s = \rho \sin V$$

What every symbol means

Symbol	Meaning	Units
x_s, y_s, z_s	point coordinates in the *sensor* frame (subscript s)	m
ρ	measured range	m
H	horizontal angle (azimuth in the sensor), measured from the sensor's x-axis toward its y-axis	deg/rad
V	vertical angle above the sensor's horizontal plane (elevation). Note: a total station's *zenith* angle Z_a is $90° - V$	deg/rad
$\rho \cos V$	the horizontal part of the range (its shadow on the floor)	m

ENGINEER'S VIEW

Read the equation in two moves. **First tilt:** the vertical angle splits the range into a horizontal part $\rho \cos V$ and a height $\rho \sin V$. **Then turn:** the horizontal angle splits the horizontal part into x and y. That is why $\cos V$ appears in both x and y. In a spinning scanner, H comes from the motor's angle encoder at the instant of firing, and V is the fixed elevation angle of the laser channel ("ring") that fired. Conventions differ by manufacturer (some measure H clockwise, some use zenith angles), so always check the sensor's documentation.

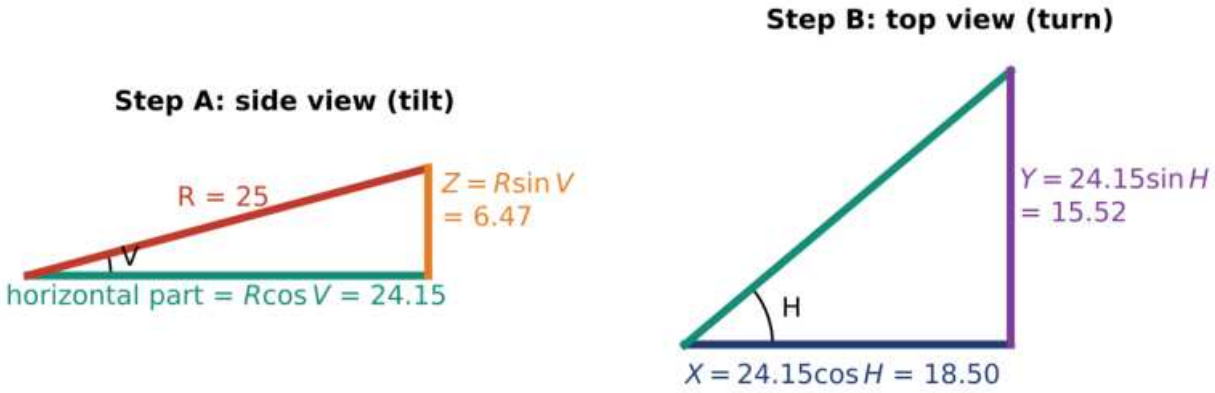


Figure 5.2: The two moves: tilt (side view), then turn (top view).

WORKED EXAMPLE — one laser return to sensor coordinates

Given: $\rho = 25.000$ m, $H = 40.0°$, $V = 15.0°$. Sensor frame: x forward, y left, z up.

Step 1 — trig values

$\cos 15° = 0.96593$, $\sin 15° = 0.25882$, $\cos 40° = 0.76604$, $\sin 40° = 0.64279$

Step 2 — horizontal part

$\rho \cos V = 25.000 \times 0.96593 = 24.148$ m

Step 3 — split it

$x_s = 24.148 \times 0.76604 = 18.499$ m; $\quad y_s = 24.148 \times 0.64279 = 15.522$ m

Step 4 — height

$z_s = 25.000 \times 0.25882 = 6.470$ m

Step 5 — check

$\sqrt{18.499^2 + 15.522^2 + 6.470^2} = \sqrt{342.21 + 240.93 + 41.86} = \sqrt{625.00} = 25.000$ ✓

Result: $(x_s, y_s, z_s) = (18.50,\ 15.52,\ 6.47)$ m.

IMPROVED FROM THE SOURCE MATERIAL

The seminar used "Y right" with a counter-clockwise H. Those two choices are inconsistent for a right-handed frame with Z up: with X forward and Z up, a right-handed Y points *left*. We use the robotics standard (ROS REP-103: x forward, y left, z up) throughout this course, and we added the length check in Step 5 — a habit worth keeping: the computed coordinates must reproduce the measured range.

5.2 Going backward: Cartesian to polar

KEY EQUATION — Inverse conversion

$$\rho = \sqrt{x^2 + y^2 + z^2}, \qquad H = \operatorname{atan2}(y, x), \qquad V = \operatorname{atan2}\left(z, \sqrt{x^2 + y^2}\right)$$

What every symbol means

Symbol	Meaning	Units
$\operatorname{atan2}(a, b)$	the angle whose tangent is a/b, placed in the correct quadrant using the signs of a and b	rad

This is what the SLAM software uses to *predict* what range and bearing the sensor should see to a known object (Module 10).

5.3 Mechanical versus solid-state scanners

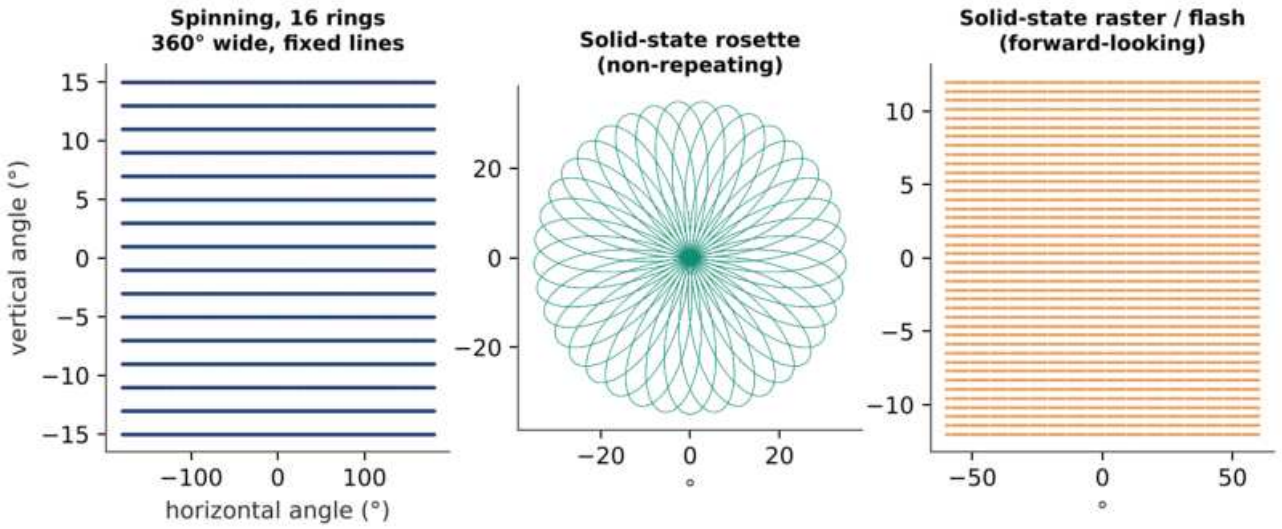


Figure 5.3: How three kinds of scanner sample the world (angles in degrees).

	Mechanical (spinning)	Solid-state
How the beam moves	a motor spins a stack of 16–128 lasers	MEMS mirror, prisms (rosette), optical phased array, or flash (no motion)
Field of view	360° horizontal, 30–90° vertical	typically 60–120° wide, forward-looking
Point pattern	regular rings	raster or non-repeating; coverage fills in over time
Strengths	full surround view, mature, well-modeled	small, light, rugged, cheap, no large moving parts
Watch-outs	motor wear, size, cost	narrower view; need several units or motion for full coverage; pattern-specific calibration

5.4 How dense will my points be?

SIMPLE APPROACH

Imagine painting a road with a paint-roller that only touches the road once every big step. If you walk faster, the stripes get further apart. If you roll faster, they get closer.

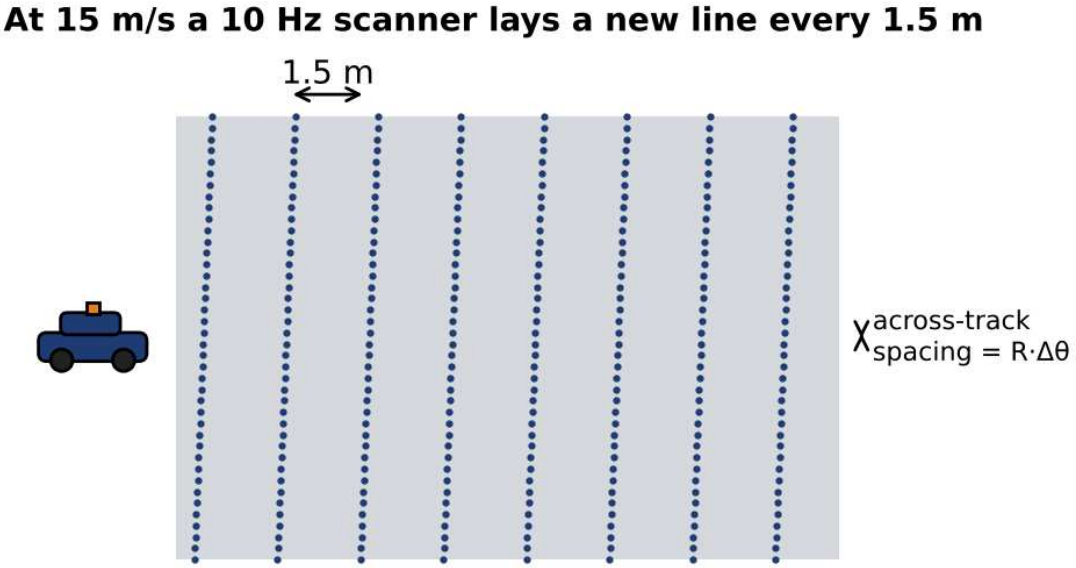


Figure 5.4: Each sweep of the scanner lays a line of points across the road.

KEY EQUATION — Point spacing and point rate

$$\Delta_{\text{along}} = \frac{v}{f_{\text{rot}}}, \qquad \Delta_{\text{across}} \approx \rho\,\Delta\theta, \qquad N_{\text{pts/s}} = N_{\text{ch}} \times N_{\text{az}} \times f_{\text{rot}}$$

What every symbol means

Symbol	Meaning	Units
Δ_{along}	spacing between scan lines in the direction of travel	m
v	platform speed	m/s
f_{rot}	rotation (scan-line) rate	Hz
Δ_{across}	spacing between neighboring points along one scan line	m
$\Delta\theta$	angular step between pulses	rad
N_{ch}	number of laser channels	—
N_{az}	azimuth steps per revolution ($360^\circ / \Delta\theta$)	—

WORKED EXAMPLE — planning a corridor survey

Given: 16-channel scanner, 0.2° horizontal step, 10 Hz, driving 15 m/s (54 km/h). Target 20 m away.

Step 1 — along-track

$\Delta_{\text{along}} = 15/10 = 1.5$ m between sweeps (for each channel)

Step 2 — across-track
$0.2^\circ = 0.003491$ rad; $\Delta_{\text{across}} = 20 \times 0.003491 = 0.070$ m

Step 3 — point rate
$N_{\text{az}} = 360/0.2 = 1800$; $N = 16 \times 1800 \times 10 = 288\,000$ points/s

Step 4 — slow down
at 5 m/s the along-track spacing becomes 0.5 m

Result: 1.5 m between lines is fine for a pavement surface but too coarse for curb lines or small utilities — drive slower, tilt the scanner, or use more channels.

SURVEYOR'S FIELD TIP

Mobile mapping heads for survey work are often mounted *tilted* so the spinning plane slices the road at an angle; then several channels hit the same strip of road at different times and along-track gaps close up. Point-density specifications should be checked on the delivered data (points per m^2 on a test patch), not assumed from the brochure.

FDOT states density in points per square foot: at least 20 points/ft^2 (about 215 points/m^2) for a Type A survey and 10 points/ft^2 (about 108 points/m^2) for Type B (Handbook Section 39.6.2; Module 19). Multiply points/ft^2 by 10.764 to get points/m^2.

TRY IT YOURSELF

5.1 Convert $\rho = 40.000$ m, $H = -30^\circ$, $V = 5^\circ$ into (x_s, y_s, z_s). Check the length.
5.2 A point is at $(3.0, 4.0, 12.0)$ m in the sensor frame. Find ρ, H and V.
5.3 A 32-channel scanner spins at 20 Hz with 0.4° steps. How many points per second? What is the along-track spacing at 10 m/s?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 163.

SOURCES FOR THIS MODULE

ROS REP-103, *Standard Units of Measure and Coordinate Conventions*; Ouster OS0 Rev8 datasheet (channels, columns, field of view).